## 10.1 Measuring an angle in radians

**Key point · Definition and results.** The unit circle is the circle with centre O and radius 1, oriented anticlockwise. One radian corresponds to an arc of length 1 on this circle. A full turn is $2\pi$ radians, a half turn $\pi$. On a circle of radius r, an arc length is $r\theta$ for a positive angle $\theta$ expressed in radians.

**Status and justification.** Definitions and proportionality between central angle and arc length. The conversion is $180° = \pi$ radians; we multiply degrees by $\frac{\pi}{180}$.

**Example 1.** $60° = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270°$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of $2\pi$ cm.

**Watch out.** The length of an arc is not that of the chord. In trigonometric calculations, check the degree or radian mode of the tool.

**Application.** Convert 45°, 120°; give the arc for radius 2 m and angle $\frac{3\pi}{4}$.

## 10.2 Wrapping and the image of a real number

**Key point · Definition and results.** To each real number x corresponds a point M of the circle, obtained by an oriented path of length x starting from I(1;0). Two real numbers have the same image if their difference is an integer multiple of $2\pi$. A negative number indicates a path in the clockwise direction.

**Status and justification.** Adding or removing a whole number of turns keeps the end point. Conversely, returning to the same point requires a whole number of turns.

**Example 2.** $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: the point $(0, -1)$.

**Watch out.** Adding $\pi$ gives the opposite point, not the same point.

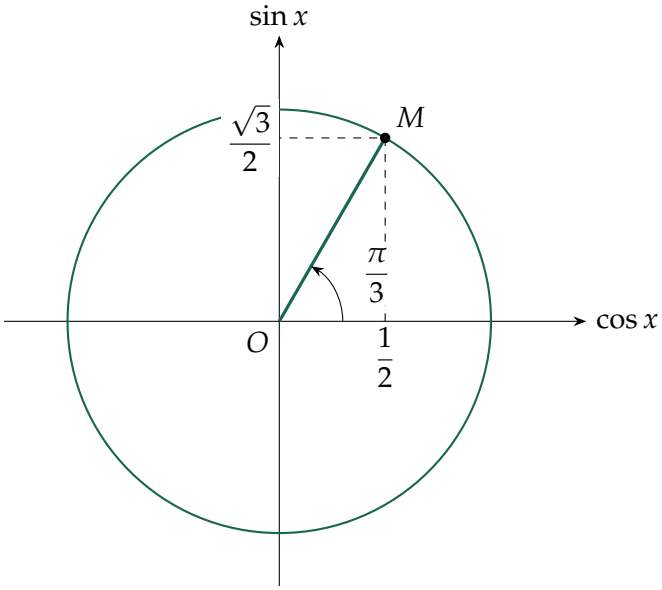
**Application.** Place the images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ by reducing them to within one turn.

## 10.3 Sine, cosine and standard values

**Key point · Definition and results.** If M is the image of x, $M = (\cos x, \sin x)$. So $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$. For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosines are $1, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, 0$ and the sines $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

**Status and justification.** At $\frac{\pi}{4}$, the two coordinates are equal and positive; Pythagoras gives $2a^2 = 1$, so $a = \frac{\sqrt{2}}{2}$. An equilateral triangle of side 1, cut by its altitude, has half-base $\frac{1}{2}$ and height $\frac{\sqrt{3}}{2}$: $\cos(\frac{\pi}{3}) = \frac{1}{2}$, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. The complementary angle gives $\frac{\pi}{6}$.

**Example 3.** The point at angle $\frac{\pi}{3}$ is $(\frac{1}{2}, \frac{\sqrt{3}}{2})$. The scalar product of two unit vectors forming this angle equals its cosine, so $\frac{1}{2}$. The geometry chapters thus connect to the circle.




For the angle $\frac{\pi}{3}$, the projections of M give its cosine and sine directly. The fractions are placed outside the lines of the drawing.

**Watch out.** The cosine is the x-coordinate, the sine the y-coordinate. The identity gives an absolute value: the quadrant determines the sign.

**Application.** If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

## 10.4 Related angles and symmetries

**Key point · Definition and results.** The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Adding $2k\pi$, k an integer, changes neither sine nor cosine.

**Status and justification.** Each equality can be read from the coordinates of the symmetric points: horizontal axis for $-x$, vertical axis for $\pi - x$, centre O for $\pi + x$. This is not a catalogue without justification.

**Example 4.** $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry from $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

**Watch out.** Two angles with the same cosine do not necessarily have the same sine. Reduce the angle, then read the signs.

**Application.** Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$.

## 10.5 Approximating pi and putting it to use

**Key point · Definition and results.** A regular inscribed polygon with n sides in the unit circle has half-perimeter $n\sin(\frac{\pi}{n})$; a circumscribed polygon has half-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ here. Archimedes' method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi in advance.

**Status and justification.** The inscribed perimeter is less than that of the circle and the circumscribed one greater: geometric bounds accepted. If c is the chord of one side, let $h = \sqrt{1 - \frac{c^2}{4}}$; the chord of the half-angle is $\frac{c}{\sqrt{2(1+h)}}$. This formula comes from Pythagoras and avoids a subtraction of nearby real numbers.

**Example 5.** For the inscribed hexagon, c=1 and half-perimeter 3. The circumscribed one gives $2\sqrt{3} \approx 3.4641$. By doubling the sides, we tighten the bounds. The supplied program starts from n=6 and c=1.

**Watch out.** Using a sine function directly with pi already known would check a calculation, but would not construct an approximation independent of pi.

**Application.** For n=6 and c=1, calculate the new chord after doubling.